Workflow and architecture patterns for working together 1: Workflows

Jacob Archambault

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Outline

1. Current challenges and their canonical solutions

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2. DevOps 201
    - Communication
        - Conway's Law
        - The Mythical Man Month
    - Throughput
        - Just-in-time manufacturing
        - Goldratt's theory of constraints
        - DevOps Research and Assessment (DORA)

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1967: Conway's law

'Organizations which design systems [...] are constrained to produce designs which are copies of the communication structures of these organizations.' - Melvin Conway, 'How do Committees Invent?' Datamation, 1967

Conway's law: examples

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- front-end [devs] business layer [backend devs] monolithic database [DBA team]
- a web api controller [manager] delegates most business logic to business classes [developers] which are part of the same in-memory process [team], while serving as a single entry-point for wider cross-network [cross-team] communication.

Conway's law: more examples

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    - separate the codebases for improved velocity
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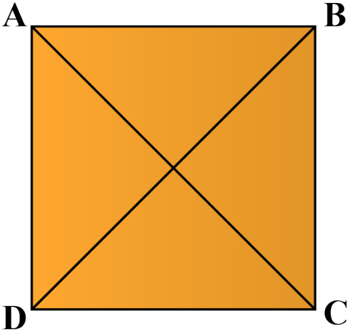
    - branching-based strategies for team management compromise between the above approaches without their benefits

1975: The Mythical Man Month, Fred Brooks

- number of direct communication paths for n individuals$=$ $n(n-1)/2$



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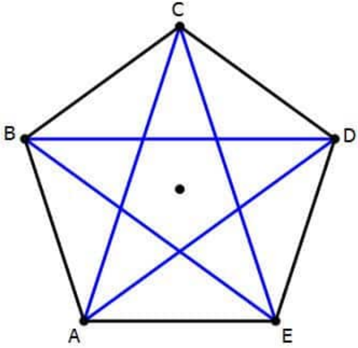

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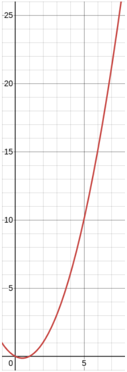
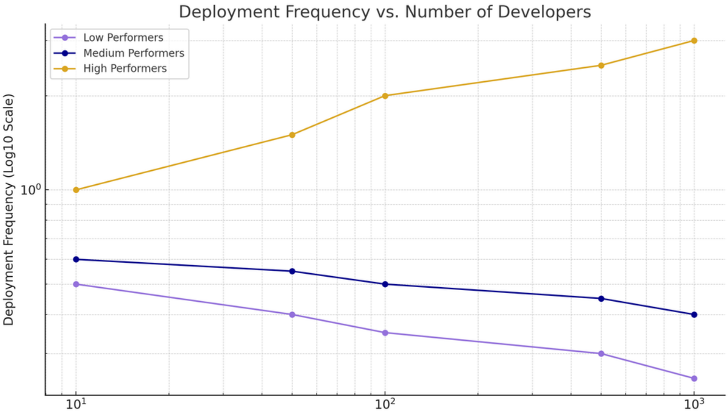


The Mythical Man Month, Fred Brooks (cont.)

- corollary: adding more people to a project can lead not only to diminishing returns on delivery speed, but to objectively less work being completed



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Just-in-time manufacturing and Goldratt's theory of constraints

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Just-in-time manufacturing and Goldratt's theory of constraints

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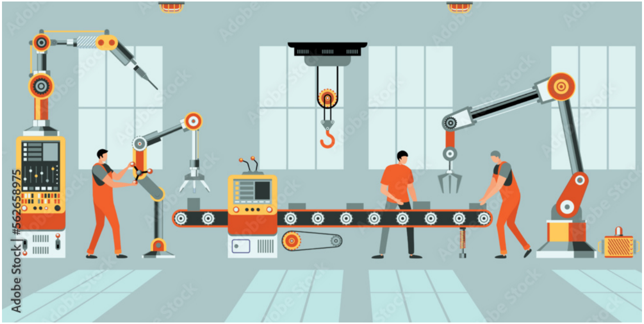
- Remove bottlenecks

Goldratt's theory of constraints (cont.)



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- Uses *cluster analysis* to discover groupings - has no prior understanding of what counts as good or bad.

The State of DevOps Report: throughput metrics

- lead time for changes

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- lead time for changes
- deployment frequency

The State of DevOps Report: stability metrics

- change failure rate

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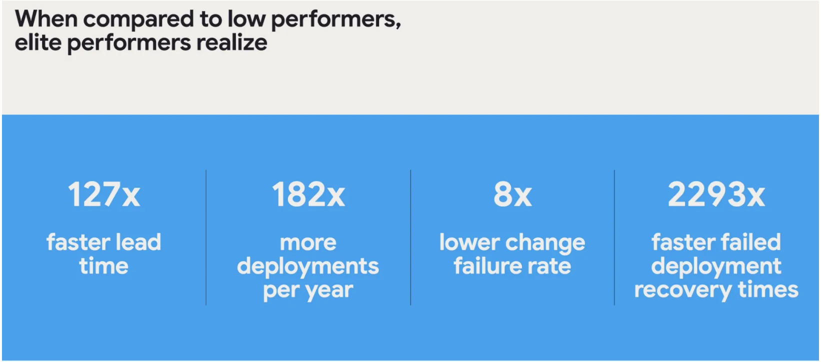
- mean time to restore

The State of DevOps Report: results

'Astonishingly, these results demonstrate that there is no trade-off between improving performance and achieving higher levels of stability and quality. Rather, high performers do better at all of these measures. This is precisely what the Agile and Lean movements predict, but much dogma in our industry still rests on the false assumption that moving faster means trading off against other performance goals, rather than enabling and reinforcing them.' - Nicole Forsgren, Jez Humble, and Gene Kim, Accelerate (2017), p. 14

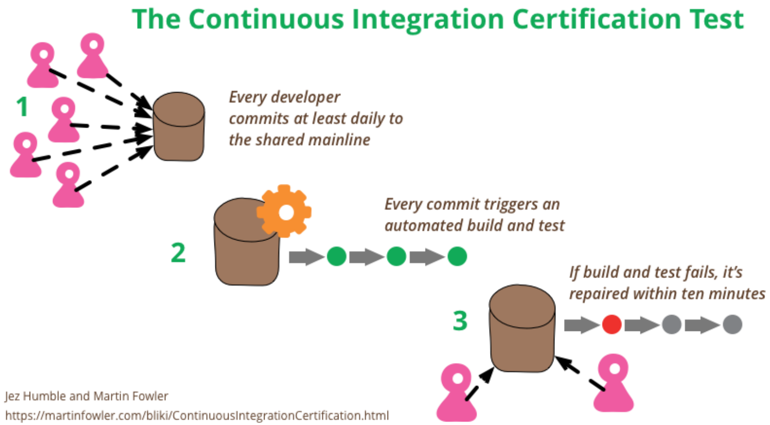
The State of DevOps Report: results



- 2024 State of DevOps Report

Outline

Defining continuous integration



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3. Branching and continuous integration
   - Definition
   - CI Branching patterns
   - Trade-offs
   - Enablers
4. Communication and throughput anti-patterns and solutions
   - Communication anti-pattern: local optimization

Continuous integration branching patterns

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Continuous integration: trade-offs

Continuous Integration applies a different trigger for integration - you integrate whenever you've made a hunk of progress on the feature and your branch is still healthy. There's no expectation that the feature be complete, just that there's been a worthwhile amount of changes to the codebase. The rule of thumb is that "everyone commits to the mainline every day", or more precisely: you should never have more than a day's work sitting unintegrated in your local repository.

- Martin Fowler, "Branching Patterns"

Feedback comes late Too late to change anything substantial	**Feedback comes in real time** As you're writing code or even before
Low quality feedback Through comments, lacking nuance	**Better quality feedback** Through face-to-face discussion
"My code" Person owns area of code; key-person dependencies	**"Our code"** Collective code ownership
Individual coding styles Harder to have standards	**Team coding style** Easier to establish standards when pairing
Infrequent integration Only integrate at the end of story/feature	**Continuous Integration** Integrate as soon as code is pushed
Easy to ignore a failing build Because it runs on an isolated branch	**We get used to not breaking things** Broken build on master is the team top priority
Dread large refactorings Because they are likely to cause merge conflicts	**Easier to tackle large refactorings** As the rest of the team is always up to date
People work in isolation Harder to spot if someone needs help	**Visibility of what everyone is doing** Easier to spot if someone needs help

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Continuous integration: Enablers

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- punishes untracked work (e.g. thorough code reviews/dev testing, helping blocked colleagues)

Solution

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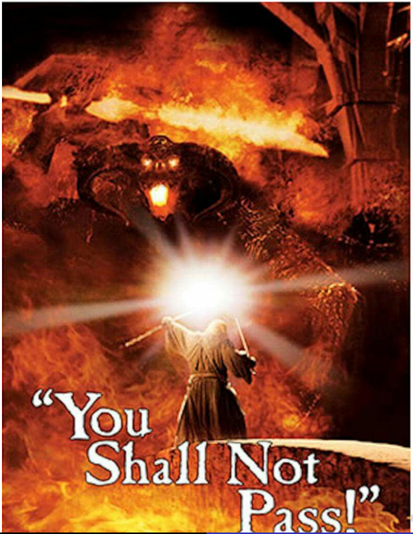

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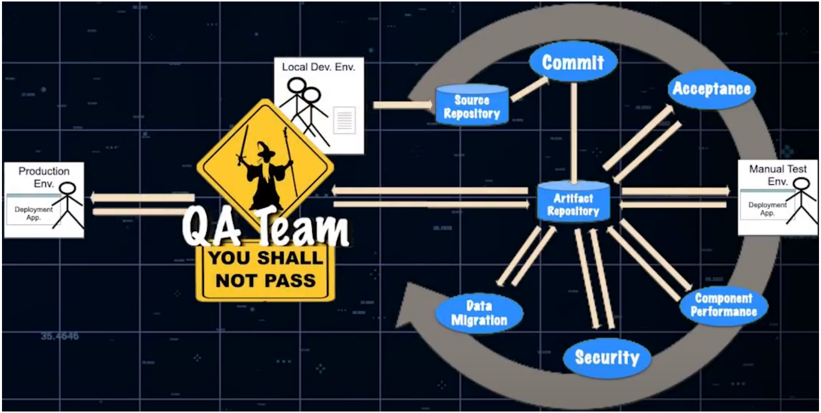

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- stop people from waiting on each other
- work in small batches
- That's mostly it

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'DevOps is whatever you do to bridge friction created by silos, and all the rest is engineering' - Patrick Debois, Puppet State of DevOps report 2021

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